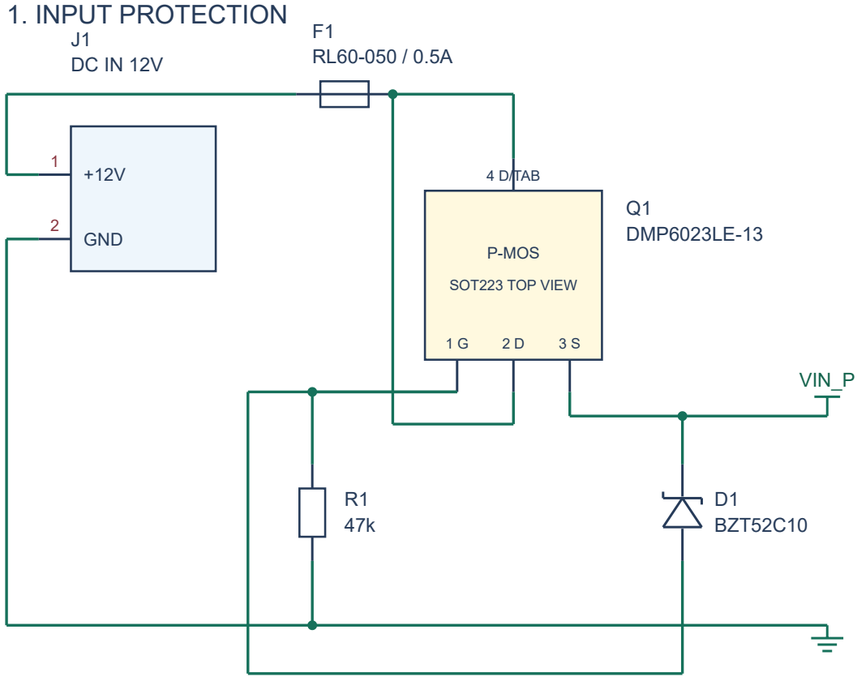
AQUARIUM ATO / POWER & PUMP

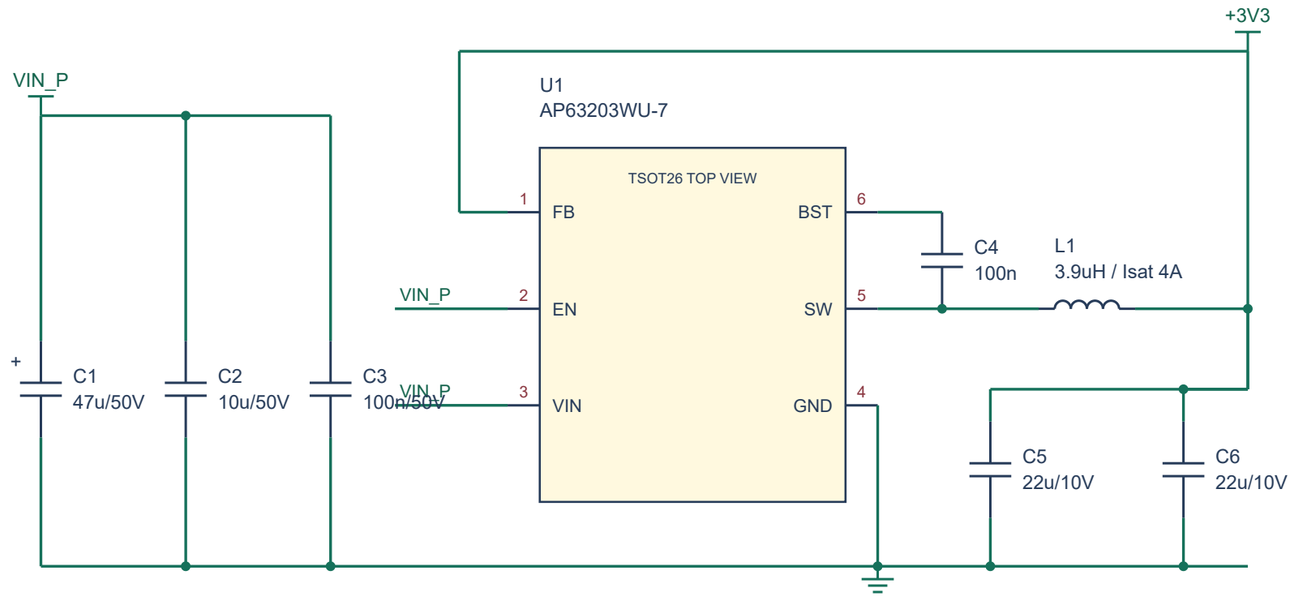
V1.0 | 12 V DC input / 3.3 V logic | 2026-10-13

1. INPUT PROTECTION



D1: cathode -> Q1 source; anode -> gate.

2. 12 V -> 3.3 V / BUCK REGULATOR

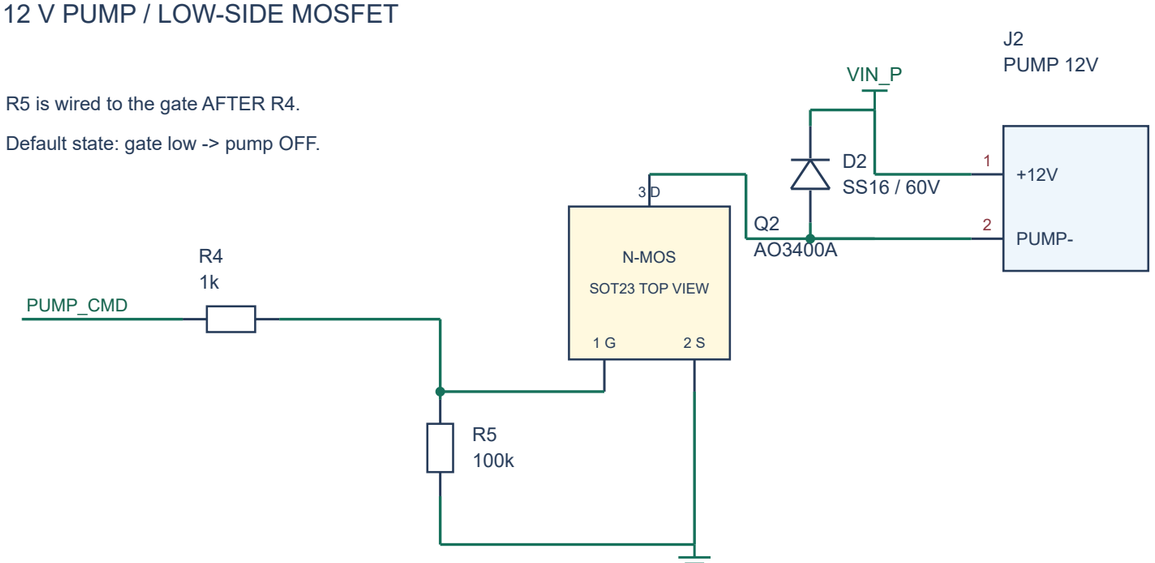


C2/C5/C6: X7R. C2/C3 close to VIN-GND. C4 across BST-SW. FB senses output AFTER L1.

3. 12 V PUMP / LOW-SIDE MOSFET

R5 is wired to the gate AFTER R4.

Default state: gate low -> pump OFF.



DESIGN / LAYOUT NOTES

01 Keep the VIN-C2-U1-GND switching loop short.

02 Place C4 at U1 pins 6 and 5; keep SW copper compact.

03 Route FB away from SW, L1 and pump return current.

04 Place D2 near J2/Q2 and use a short pump return loop.

05 VIN_P supplies both the buck converter and the pump.

06 F1 remains 0.5 A from V0.9: check pump startup current.

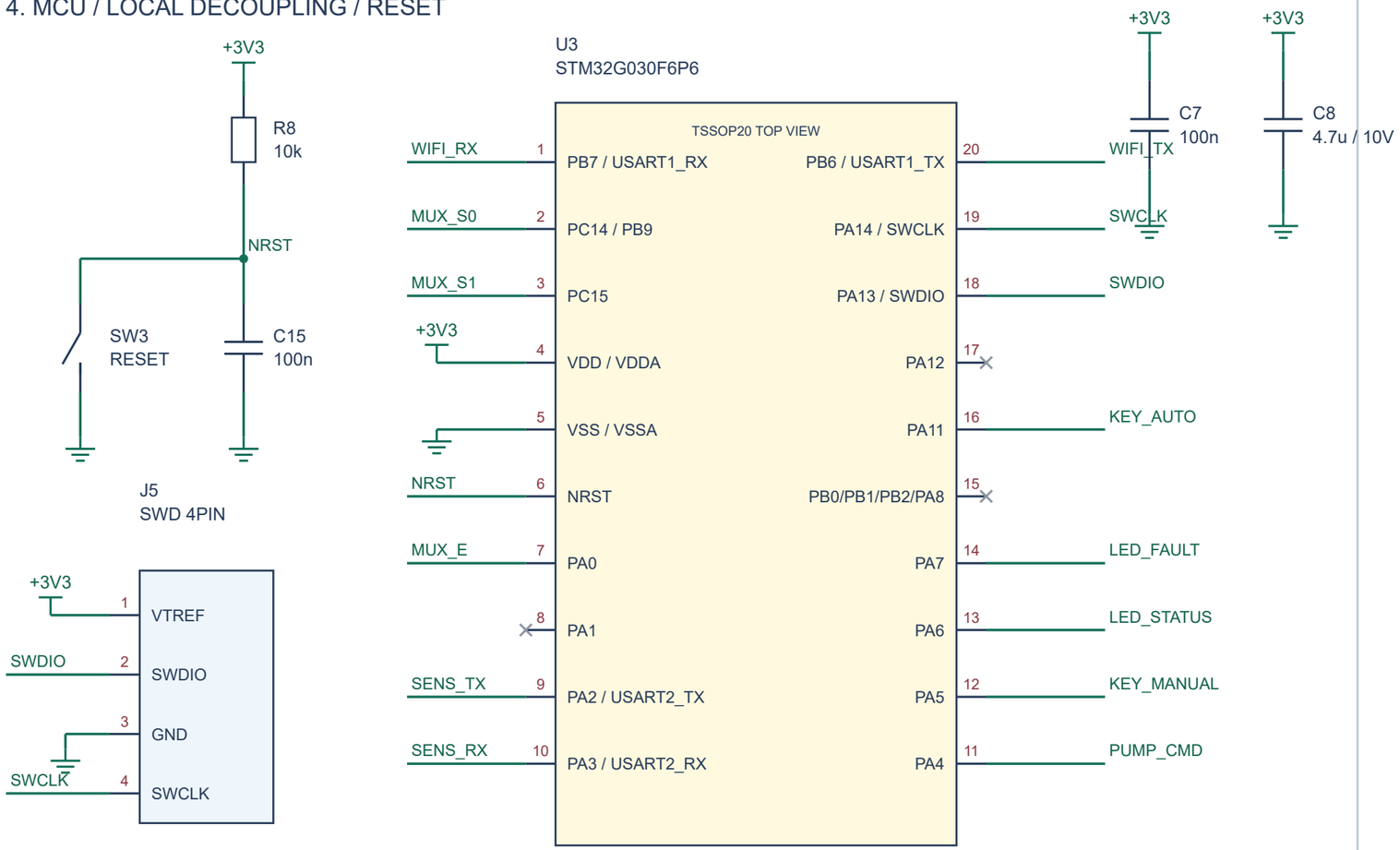
07 This revision reorganizes wiring; original values retained.

08 PCB footprints / load-current validation are not included.

AQUARIUM ATO / CONTROLLER & USER I/O

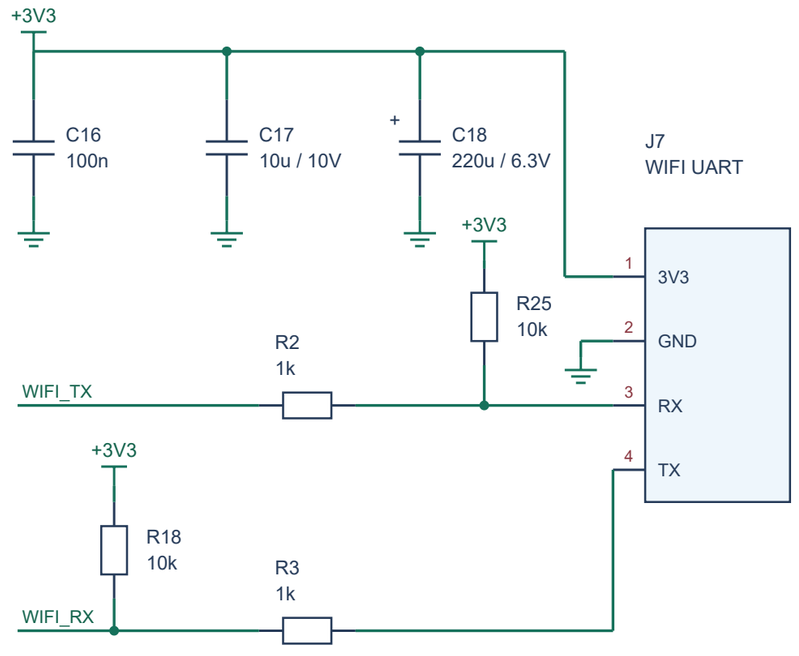
V1.0 | 12 V DC input / 3.3 V logic | 2026-10-13

4. MCU / LOCAL DECOUPLING / RESET



U3 symbol follows the TSSOP20 physical top-view pin order. Unused pins are marked X.

5. WIFI / 3.3 V UART ONLY



RX / TX on J7 are named from the MODULE side.

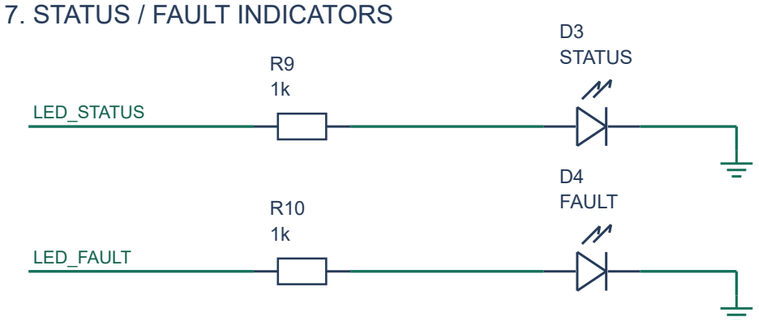
R18: MCU RX idle-high. R25: module RX idle-high.

C16/C17/C18 connect directly across J7 supply.

6. BUTTONS



7. STATUS / FAULT INDICATORS

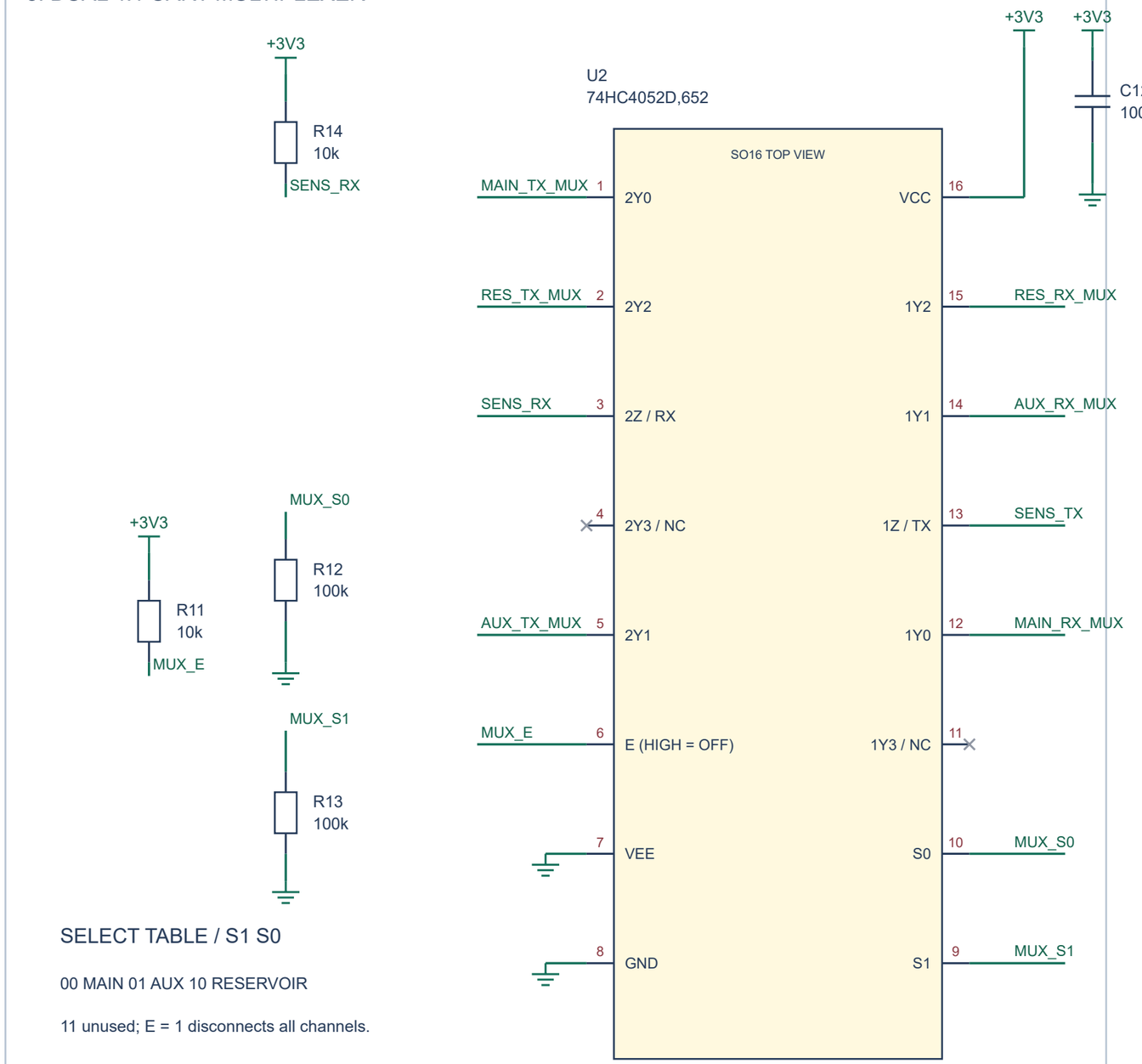


Connected dots = joined nets. Crossing lines without a dot are NOT connected. Same net names connect across sheets.

AQUARIUM ATO / SENSOR MULTIPLEXER & CONNECTORS

V1.0 | 12 V DC input / 3.3 V logic | 2026-10-13

8. DUAL 4:1 UART MULTIPLEXER



SELECT TABLE / S1 S0

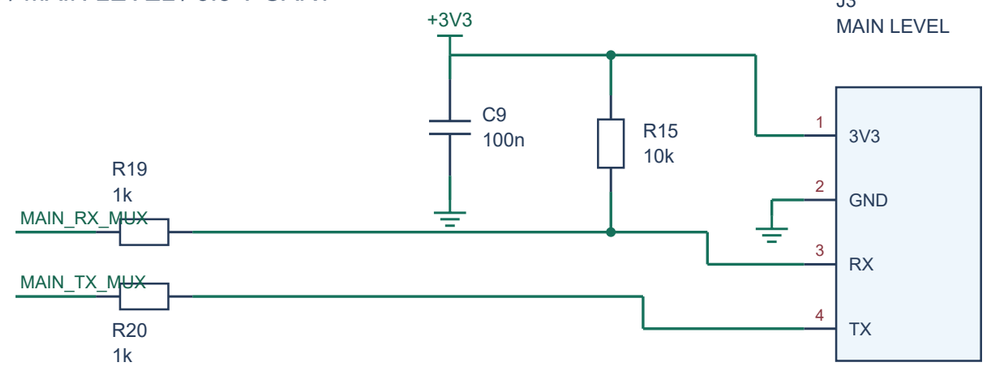
00 MAIN 01 AUX 10 RESERVOIR

11 unused; E = 1 disconnects all channels.

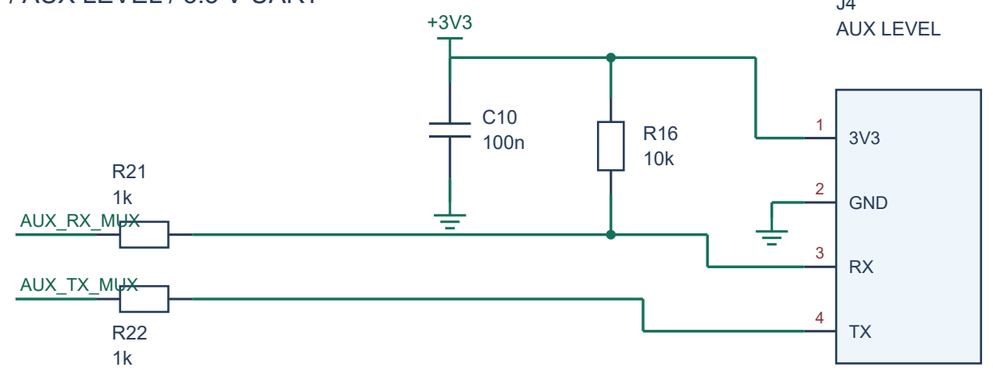
Power-up: R11 holds E high; R12/R13 hold S0/S1 low.

Disable E before changing the selected UART channel.

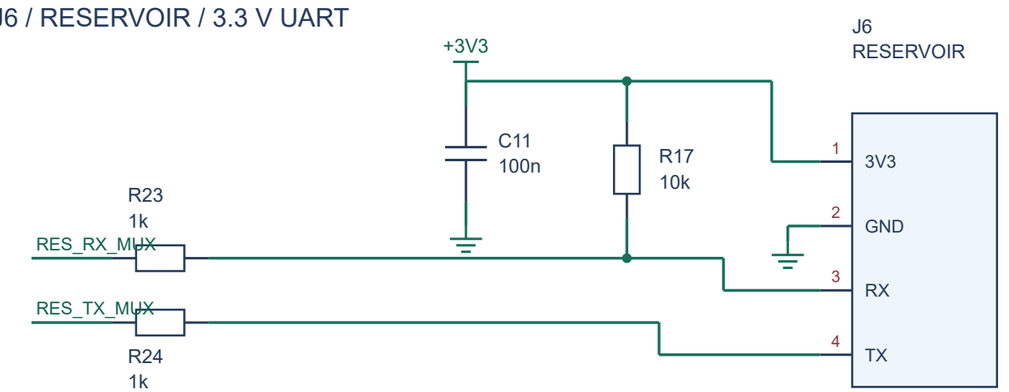
J3 / MAIN LEVEL / 3.3 V UART



J4 / AUX LEVEL / 3.3 V UART



J6 / RESERVOIR / 3.3 V UART



Connector RX/TX are from the SENSOR side. Each RX pull-up is after its series resistor.